

Reverse GCC & CLANG

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КОМПИЛЯТОРЫ

GCC, CLANG

- под две архитектуры 32, 64;
- оптимизация O1, O3

ИНСТРУМЕНТЫ

RADARE2

aaa - анализ исполняемого файла

ia - просмотр импортов и экспортов

iS entropy - подсчет энтропии

s main - переход в main

V - перейти в визуальный режим 

s 0x00000000 - перейти на конкретный адрес

GHIDRA

декомпиляция

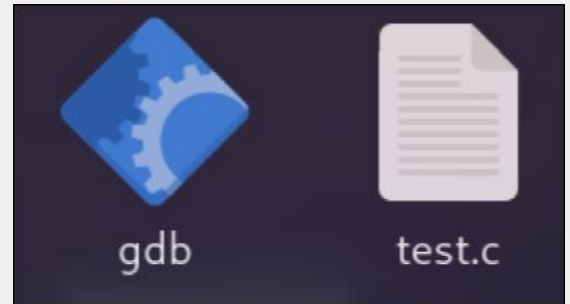
Исполняемые файлы

```
1  #include <stdio.h>
2
3
4  void f (int a)
5  {
6      switch (a) {
7      case 0:
8          printf (format: "zero\n");
9          break;
10
11      case 1:
12          printf(format: "one\n");
13          break;
14
15      case 2:
16          printf(format: "two\n");
17          break;
18
19      default:
20          printf(format: "something unknown\n");
21      }
22 }
23
24
25 int main(void)
26 {
27     f(a: 2);
28     return 0;
29 }
30
```

14676	дек 11 23:26	clang_32_01.out
14676	дек 11 23:26	clang_32_03.out
15456	дек 11 23:25	clang_64_01.out
15456	дек 11 23:25	clang_64_03.out
14640	дек 11 23:37	gcc_32_01.out
14604	дек 11 23:37	gcc_32_03.out
15440	дек 11 23:34	gcc_64_01.out
15440	дек 11 23:34	gcc_64_03.out

Исполняемые файл

```
1  #include <stdio.h>
2
3
4  void f (int a)
5  {
6      switch (a) {
7      case 0:
8          printf (format: "zero\n");
9          break;
10
11      case 1:
12          printf(format: "one\n");
13          break;
14
15      case 2:
16          printf(format: "two\n");
17          break;
18
19      default:
20          printf(format: "something unknown\n");
21      }
22  }
23
24
25  int main(void)
26  {
27      f(a: 2);
28      return 0;
29  }
30
```



GDB

Breakpoint - break

```
(gdb) break 5  
Breakpoint 1 at 0x1144: file test.c, line 5.
```

Запуск - run

```
(gdb) run  
Starting program: /home/vedroid/Reverse/gdb
```

Аргументы функции - info args

```
(gdb) info args  
a = 2
```

Посмотреть код - list

```
(gdb) list  
1      #include <stdio.h>  
2  
3      void f (int a)  
4      {  
5          switch (a) {  
6              case 0:  
7                  printf ("zero\n");  
8                  break;  
9  
10             case 1:
```

Перейти дальше - next

```
(gdb) next  
15          printf("two\n");
```

CLANG32~01

Radare2

[Imports]							
nth	vaddr	bind	type	lib name			
1	0x00001040	GLOBAL	FUNC	__libc_start_main			
2	-----	WEAK	NOTYPE	_ITM_deregisterTMCloneTable			
3	-----	WEAK	FUNC	__cxa_finalize			
4	0x00001050	GLOBAL	FUNC	puts			
5	-----	WEAK	NOTYPE	__gmon_start__			
6	-----	WEAK	NOTYPE	_ITM_registerTMCloneTable			
[Exports]							
nth	paddr	vaddr	bind	type	size lib name	demangled	
7	0x00002004	0x00002004	GLOBAL	OBJ	4	_IO_stdin_used	
8	0x00001090	0x00001090	GLOBAL	FUNC	4	__x86.get_pc_thunk.bx	
10	-----	0x00004010	GLOBAL	NOTYPE	0	_edata	
11	0x000011f8	0x000011f8	GLOBAL	FUNC	0	_fini	
12	0x00001190	0x00001190	GLOBAL	FUNC	53	f	
13	0x00001189	0x00001189	GLOBAL	FUNC	0	__x86.get_pc_thunk.dx	
15	0x00003008	0x00004008	GLOBAL	NOTYPE	0	__data_start	
18	0x0000300c	0x0000400c	GLOBAL	OBJ	0	__dso_handle	
19	0x00002004	0x00002004	GLOBAL	OBJ	4	_IO_stdin_used	
20	-----	0x00004014	GLOBAL	NOTYPE	0	_end	
21	0x00001060	0x00001060	GLOBAL	FUNC	48	_start	
22	0x00002000	0x00002000	GLOBAL	OBJ	4	_fp_hw	
23	-----	0x00004010	GLOBAL	NOTYPE	0	__bss_start	
24	0x000011d0	0x000011d0	GLOBAL	FUNC	37	main	
25	-----	0x00004010	GLOBAL	OBJ	0	__TMC_END__	
27	0x00001000	0x00001000	GLOBAL	FUNC	0	_init	

импорты и экспорты

nth	paddr	size	vaddr	vsize	perm	entropy	type	name
0	0x00000000	0x0	0x00000000	0x0	----		NULL	
1	0x000001b4	0x13	0x000001b4	0x13	-r--	3.68188080	PROGBITS	.interp
2	0x000001c8	0x24	0x000001c8	0x24	-r--	4.13605609	NOTE	.note.gnu.build-id
3	0x000001ec	0x1c	0x000001ec	0x1c	-r--	2.25425248	NOTE	.note.gnu.property
4	0x00000208	0x20	0x00000208	0x20	-r--	1.41268789	NOTE	.note.ABI-tag
5	0x00000228	0x20	0x00000228	0x20	-r--	1.99254169	GNU_HASH	.gnu.hash
6	0x00000248	0x80	0x00000248	0x80	-r--	1.04726900	DYNSYM	.dynsym
7	0x000002c8	0xa6	0x000002c8	0xa6	-r--	4.70680756	STRTAB	.dynstr
8	0x0000036e	0x10	0x0000036e	0x10	-r--	1.71691719	GNU_VERSION	.gnu.version
9	0x00000380	0x40	0x00000380	0x40	-r--	2.44324982	GNU_VERNEED	.gnu.version_r
10	0x000003c0	0x58	0x000003c0	0x58	-r--	2.46199868	REL	.rel.dyn
11	0x00000418	0x10	0x00000418	0x10	-r--	1.84191719	REL	.rel.plt
12	0x00001000	0x24	0x00001000	0x24	-r-x	4.29187943	PROGBITS	.init
13	0x00001030	0x30	0x00001030	0x30	-r-x	2.50129389	PROGBITS	.plt
14	0x00001060	0x195	0x00001060	0x195	-r-x	5.49538799	PROGBITS	.text
15	0x000011f8	0x18	0x000011f8	0x18	-r-x	4.16829583	PROGBITS	.fini
16	0x00002000	0x27	0x00002000	0x27	-r--	3.73403059	PROGBITS	.rodata
17	0x00002028	0x2c	0x00002028	0x2c	-r--	3.11718957	PROGBITS	.eh_frame_hdr
18	0x00002054	0xa4	0x00002054	0xa4	-r--	4.09094744	PROGBITS	.eh_frame
19	0x00002edc	0x4	0x00003edc	0x4	-rw-	1.50000000	INIT_ARRAY	.init_array
20	0x00002ee0	0x4	0x00003ee0	0x4	-rw-	1.50000000	FINI_ARRAY	.fini_array
21	0x00002ee4	0xc	0x00003ee4	0xc	-rw-	1.89624063	PROGBITS	.data.rel.ro
22	0x00002ef0	0xf0	0x00003ef0	0xf0	-rw-	2.54918356	DYNAMIC	.dynamic
23	0x00002fe0	0x14	0x00003fe0	0x14	-rw-	0.56899559	PROGBITS	.got
24	0x00002ffa	0x14	0x00003ffa	0x14	-rw-	1.55677965	PROGBITS	.got.plt
25	0x00003008	0x8	0x00004008	0x8	-rw-	1.06127812	PROGBITS	.data
26	0x00003010	0x0	0x00004010	0x4	-rw-		NOBITS	.bss

энтропия

CLANG32~01

Ghidra

```
int main (int argc, char **argv, char **envp);  
0x000011d0      push ebx  
0x000011d1      sub esp, 8  
0x000011d4      call 0x11d9  
; CALL XREF from main @ 0x11d4(x)  
0x000011d9      pop ebx  
0x000011da      add ebx, 0x2e1b  
0x000011e0      lea eax, [ebx - 0x1fec]  
0x000011e6      mov dword [esp], eax  
0x000011e9      call sym.imp.puts  
0x000011ee      xor eax, eax  
0x000011f0      add esp, 8  
0x000011f3      pop ebx  
0x000011f4      ret
```

дизасемблирование main

```
2 int main(void)  
3  
4 {  
5     puts("two");  
6     return 0;  
7 }
```

декомпилирование

CLANG32~03

Radare2

[Imports]					
nth	vaddr	bind	type	lib	name
1	0x00001040	GLOBAL	FUNC	__libc_start_main	
2	-----	WEAK	NOTYPE	_ITM_deregisterTMCloneTable	
3	-----	WEAK	FUNC	__cxa_finalize	
4	0x00001050	GLOBAL	FUNC	puts	
5	-----	WEAK	NOTYPE	__gmon_start__	
6	-----	WEAK	NOTYPE	_ITM_registerTMCloneTable	
[Exports]					
nth	paddr	vaddr	bind	type	size lib name demangled
7	0x00002004	0x00002004	GLOBAL	OBJ	4 _IO_stdin_used
8	0x00001090	0x00001090	GLOBAL	FUNC	4 __x86.get_pc_thunk.bx
10	-----	0x00004010	GLOBAL	NOTYPE	0 _edata
11	0x000011f8	0x000011f8	GLOBAL	FUNC	0 _fini
12	0x00001190	0x00001190	GLOBAL	FUNC	53 f
13	0x00001189	0x00001189	GLOBAL	FUNC	0 __x86.get_pc_thunk.dx
15	0x00003008	0x00004008	GLOBAL	NOTYPE	0 __data_start
18	0x0000300c	0x0000400c	GLOBAL	OBJ	0 __dso_handle
19	0x00002004	0x00002004	GLOBAL	OBJ	4 _IO_stdin_used
20	-----	0x00004014	GLOBAL	NOTYPE	0 _end
21	0x00001060	0x00001060	GLOBAL	FUNC	48 _start
22	0x00002000	0x00002000	GLOBAL	OBJ	4 _fp_hw
23	-----	0x00004010	GLOBAL	NOTYPE	0 __bss_start
24	0x000011d0	0x000011d0	GLOBAL	FUNC	37 main
25	-----	0x00004010	GLOBAL	OBJ	0 __TMC_END__
27	0x00001000	0x00001000	GLOBAL	FUNC	0 _init

импорты и экспорты

nth	paddr	size	vaddr	vsize	perm	entropy	type	name
0	0x00000000	0x0	0x00000000	0x0	----		NULL	
1	0x000001b4	0x13	0x000001b4	0x13	-r--	3.68188080	PROGBITS	.interp
2	0x000001c8	0x24	0x000001c8	0x24	-r--	4.13605609	NOTE	.note.gnu.build-id
3	0x000001ec	0x1c	0x000001ec	0x1c	-r--	2.25425248	NOTE	.note.gnu.property
4	0x00000208	0x20	0x00000208	0x20	-r--	1.41268789	NOTE	.note.ABI-tag
5	0x00000228	0x20	0x00000228	0x20	-r--	1.99254169	GNU_HASH	.gnu.hash
6	0x00000248	0x80	0x00000248	0x80	-r--	1.04726900	DYNSYM	.dynsym
7	0x000002c8	0xa6	0x000002c8	0xa6	-r--	4.70680756	STRTAB	.dynstr
8	0x0000036e	0x10	0x0000036e	0x10	-r--	1.71691719	GNU_VERSION	.gnu.version
9	0x00000380	0x40	0x00000380	0x40	-r--	2.44324982	GNU_VERNEED	.gnu.version_r
10	0x000003c0	0x58	0x000003c0	0x58	-r--	2.46199868	REL	.rel.dyn
11	0x00000418	0x10	0x00000418	0x10	-r--	1.84191719	REL	.rel.plt
12	0x00001000	0x24	0x00001000	0x24	-r-x	4.29187943	PROGBITS	.init
13	0x00001030	0x30	0x00001030	0x30	-r-x	2.50129389	PROGBITS	.plt
14	0x00001060	0x195	0x00001060	0x195	-r-x	5.49538799	PROGBITS	.text
15	0x000011f8	0x18	0x000011f8	0x18	-r-x	4.16829583	PROGBITS	.fini
16	0x00002000	0x27	0x00002000	0x27	-r--	3.73403059	PROGBITS	.rodata
17	0x00002028	0x2c	0x00002028	0x2c	-r--	3.11718957	PROGBITS	.eh_frame_hdr
18	0x00002054	0xa4	0x00002054	0xa4	-r--	4.09094744	PROGBITS	.eh_frame
19	0x00002edc	0x4	0x00003edc	0x4	-rw-	1.50000000	INIT_ARRAY	.init_array
20	0x00002ee0	0x4	0x00003ee0	0x4	-rw-	1.50000000	FINI_ARRAY	.fini_array
21	0x00002ee4	0xc	0x00003ee4	0xc	-rw-	1.89624063	PROGBITS	.data.rel.ro
22	0x00002ef0	0xf0	0x00003ef0	0xf0	-rw-	2.54918356	DYNAMIC	.dynamic
23	0x00002fe0	0x14	0x00003fe0	0x14	-rw-	0.56899559	PROGBITS	.got
24	0x00002ff4	0x14	0x00003ff4	0x14	-rw-	1.55677965	PROGBITS	.got.plt
25	0x00003008	0x8	0x00004008	0x8	-rw-	1.06127812	PROGBITS	.data
26	0x00003010	0x0	0x00004010	0x4	-rw-		NOBITS	.bss

энтропия

CLANG32~03

Ghidra

```
int main (int argc, char **argv, char **envp);
0x000011d0      push ebx
0x000011d1      sub esp, 8
0x000011d4      call 0x11d9
; CALL XREF from main @ 0x11d4(x)
0x000011d9      pop ebx
0x000011da      add ebx, 0x2e1b
0x000011e0      lea eax, [ebx - 0x1fec]
0x000011e6      mov dword [esp], eax
0x000011e9      call sym.imp.puts
0x000011ee      xor eax, eax
0x000011f0      add esp, 8
0x000011f3      pop ebx
0x000011f4      ret
```

дизасемблирование main

```
2 int main(void)
3
4 {
5     puts("two");
6     return 0;
7 }
```

декомпилирование

CLANG64~01

Radare2

[Imports]						
nth	vaddr	bind	type	lib	name	
1	-----	GLOBAL	FUNC		__libc_start_main	
2	-----	WEAK	NOTYPE		_ITM_deregisterTMCloneTable	
3	0x00001030	GLOBAL	FUNC		puts	
4	-----	WEAK	NOTYPE		__gmon_start__	
5	-----	WEAK	NOTYPE		_ITM_registerTMCloneTable	
6	-----	WEAK	FUNC		__cxa_finalize	
[Exports]						
nth	paddr	vaddr	bind	type	size	lib name demangled
10	-----	0x00004018	GLOBAL	NOTYPE	0	_edata
11	0x00001184	0x00001184	GLOBAL	FUNC	0	_fini
12	0x00001140	0x00001140	GLOBAL	FUNC	39	f
13	0x00003008	0x00004008	GLOBAL	NOTYPE	0	__data_start
15	0x00003010	0x00004010	GLOBAL	OBJ	0	__dso_handle
16	0x00002000	0x00002000	GLOBAL	OBJ	4	_IO_stdin_used
17	-----	0x00004020	GLOBAL	NOTYPE	0	_end
18	0x00001040	0x00001040	GLOBAL	FUNC	38	_start
19	-----	0x00004018	GLOBAL	NOTYPE	0	__bss_start
20	0x00001170	0x00001170	GLOBAL	FUNC	17	main

импорты и экспорты

nth	paddr	size	vaddr	vsize	perm	entropy	type	name
0	0x00000000	0x0	0x00000000	0x0	----		NULL	
1	0x00000318	0x1c	0x00000318	0x1c	-r--	3.94075983	PROGBITS	.interp
2	0x00000338	0x20	0x00000338	0x20	-r--	2.05504169	NOTE	.note.gnu.property
3	0x00000358	0x24	0x00000358	0x24	-r--	4.19161164	NOTE	.note.gnu.build-id
4	0x0000037c	0x20	0x0000037c	0x20	-r--	1.41268789	NOTE	.note.ABI-tag
5	0x000003a0	0x1c	0x000003a0	0x1c	-r--	0.49123734	GNU_HASH	.gnu.hash
6	0x000003c0	0xa8	0x000003c0	0xa8	-r--	0.57518865	DYNSYM	.dynsym
7	0x00000468	0x8d	0x00000468	0x8d	-r--	4.64227179	STRTAB	.dynstr
8	0x000004f6	0xe	0x000004f6	0xe	-r--	1.61057724	GNU_VERSION	.gnu.version
9	0x00000508	0x30	0x00000508	0x30	-r--	2.30204405	GNU_VERNEED	.gnu.version_r
10	0x00000538	0xc0	0x00000538	0xc0	-r--	1.37722996	RELA	.rela.dyn
11	0x000005f8	0x18	0x000005f8	0x18	-r--	0.74168476	RELA	.rela.plt
12	0x00001000	0x1b	0x00001000	0x1b	-r-x	4.23636898	PROGBITS	.init
13	0x00001020	0x20	0x00001020	0x20	-r-x	3.05254968	PROGBITS	.plt
14	0x00001040	0x141	0x00001040	0x141	-r-x	5.19053458	PROGBITS	.text
15	0x00001184	0xd	0x00001184	0xd	-r-x	3.23890126	PROGBITS	.fini
16	0x00002000	0x30	0x00002000	0x30	-r--	3.91506713	PROGBITS	.rodata
17	0x00002030	0x2c	0x00002030	0x2c	-r--	3.07173503	PROGBITS	.eh_frame_hdr
18	0x00002060	0x88	0x00002060	0x88	-r--	3.57608496	PROGBITS	.eh_frame
19	0x00002dd0	0x8	0x00003dd0	0x8	-rw-	1.06127812	INIT_ARRAY	.init_array
20	0x00002dd8	0x8	0x00003dd8	0x8	-rw-	1.06127812	FINI_ARRAY	.fini_array
21	0x00002de0	0x1e0	0x00003de0	0x1e0	-rw-	1.44727261	DYNAMIC	.dynamic
22	0x00002fc0	0x28	0x00003fc0	0x28	-rw-	0.00000000	PROGBITS	.got
23	0x00002fe8	0x20	0x00003fe8	0x20	-rw-	0.79356444	PROGBITS	.got.plt
24	0x00003008	0x10	0x00004008	0x10	-rw-	0.66856444	PROGBITS	.data
25	0x00003018	0x0	0x00004018	0x8	-rw-		NOBITS	.bss
26	0x00003018	0x30	0x00000000	0x30	----	4.49403483	PROGBITS	.comment

энтропия

CLANG64~01

Ghidra

```
    ; DATA XREF from entry0 @ 0x1058(r)
int main (int argc, char **argv, char **envp);
0x00001170    push rax
0x00001171    lea rdi, [0x00002004]
0x00001178    call sym.imp.puts
0x0000117d    xor eax, eax
0x0000117f    pop rcx
0x00001180    ret
```

дизасемблирование main

```
2 int main(void)
3
4 {
5     puts("two");
6     return 0;
7 }
```

декомпилирование

CLANG64~03

Radare2

[Imports]				
nth	vaddr	bind	type	lib name
1	-----	GLOBAL	FUNC	__libc_start_main
2	-----	WEAK	NOTYPE	_ITM_deregisterTMCloneTable
3	0x00001030	GLOBAL	FUNC	puts
4	-----	WEAK	NOTYPE	__gmon_start__
5	-----	WEAK	NOTYPE	_ITM_registerTMCloneTable
6	-----	WEAK	FUNC	__cxa_finalize

[Exports]						
nth	paddr	vaddr	bind	type	size	lib name
10	-----	0x00004018	GLOBAL	NOTYPE	0	__edata
11	0x00001184	0x00001184	GLOBAL	FUNC	0	__fini
12	0x00001140	0x00001140	GLOBAL	FUNC	39	f
13	0x00003008	0x00004008	GLOBAL	NOTYPE	0	__data_start
15	0x00003010	0x00004010	GLOBAL	OBJ	0	__dso_handle
16	0x00002000	0x00002000	GLOBAL	OBJ	4	_IO_stdin_used
17	-----	0x00004020	GLOBAL	NOTYPE	0	__end
18	0x00001040	0x00001040	GLOBAL	FUNC	38	__start
19	-----	0x00004018	GLOBAL	NOTYPE	0	__bss_start
20	0x00001170	0x00001170	GLOBAL	FUNC	17	main

импорты и экспорты

[Sections]						
nth	paddr	size	vaddr	vsize	perm	entropy
0	0x00000000	0x0	0x00000000	0x0	----	NULL
1	0x00000318	0x1c	0x00000318	0x1c	-r--	3.94075983
2	0x00000338	0x20	0x00000338	0x20	-r--	2.05504169
3	0x00000358	0x24	0x00000358	0x24	-r--	4.19161164
4	0x0000037c	0x20	0x0000037c	0x20	-r--	1.41268789
5	0x000003a0	0x1c	0x000003a0	0x1c	-r--	0.49123734
6	0x000003c0	0xa8	0x000003c0	0xa8	-r--	0.57518865
7	0x00000468	0x8d	0x00000468	0x8d	-r--	4.64227179
8	0x000004f6	0xe	0x000004f6	0xe	-r--	1.61057724
9	0x00000508	0x30	0x00000508	0x30	-r--	2.30204405
10	0x00000538	0xc0	0x00000538	0xc0	-r--	1.37722996
11	0x000005f8	0x18	0x000005f8	0x18	-r--	0.74168476
12	0x00001000	0x1b	0x00001000	0x1b	-r-x	4.23636898
13	0x00001020	0x20	0x00001020	0x20	-r-x	3.05254968
14	0x00001040	0x141	0x00001040	0x141	-r-x	5.19053458
15	0x00001184	0xd	0x00001184	0xd	-r-x	3.23890126
16	0x00002000	0x30	0x00002000	0x30	-r--	3.91506713
17	0x00002030	0x2c	0x00002030	0x2c	-r--	3.07173503
18	0x00002060	0x88	0x00002060	0x88	-r--	3.57608496
19	0x00002dd0	0x8	0x00003dd0	0x8	-rw-	1.06127812
20	0x00002dd8	0x8	0x00003dd8	0x8	-rw-	1.06127812
21	0x00002de0	0x1e0	0x00003de0	0x1e0	-rw-	1.44727261
22	0x00002fc0	0x28	0x00003fc0	0x28	-rw-	0.00000000
23	0x00002fe8	0x20	0x00003fe8	0x20	-rw-	0.79356444
24	0x00003008	0x10	0x00004008	0x10	-rw-	0.66856444

энтропия

CLANG64~03

Ghidra

```
    ; DATA XREF from entry0 @ 0x1058(r)
int main (int argc, char **argv, char **envp);
0x00001170    push rax
0x00001171    lea rdi, [0x00002004]
0x00001178    call sym.imp.puts
0x0000117d    xor eax, eax
0x0000117f    pop rcx
0x00001180    ret
```

дизасемблирование main

```
2 int main(void)
3
4 {
5     puts("two");
6     return 0;
7 }
```

декомпилирование

GCC32~01

Radare2

[Imports]							
nth	vaddr	bind	type	lib name			
1	0x00001040	GLOBAL	FUNC	__libc_start_main			
2	-----	WEAK	NOTYPE	_ITM_deregisterTMCloneTable			
3	-----	WEAK	FUNC	__cxa_finalize			
4	0x00001050	GLOBAL	FUNC	puts			
5	-----	WEAK	NOTYPE	__gmon_start__			
6	-----	WEAK	NOTYPE	_ITM_registerTMCloneTable			
[Exports]							
nth	paddr	vaddr	bind	type	size lib name	demangled	
7	0x00002004	0x00002004	GLOBAL	OBJ	4	_IO_stdin_used	
8	0x00001090	0x00001090	GLOBAL	FUNC	4	__x86.get_pc_thunk.bx	
10	-----	0x00004010	GLOBAL	NOTYPE	0	_edata	
11	0x00001250	0x00001250	GLOBAL	FUNC	0	_fini	
12	0x0000118d	0x0000118d	GLOBAL	FUNC	136	f	
13	0x00001189	0x00001189	GLOBAL	FUNC	0	__x86.get_pc_thunk.dx	
15	0x00003008	0x00004008	GLOBAL	NOTYPE	0	__data_start	
18	0x0000300c	0x0000400c	GLOBAL	OBJ	0	__dso_handle	
19	0x00002004	0x00002004	GLOBAL	OBJ	4	_IO_stdin_used	
20	-----	0x00004014	GLOBAL	NOTYPE	0	_end	
21	0x00001060	0x00001060	GLOBAL	FUNC	48	_start	
22	0x00002000	0x00002000	GLOBAL	OBJ	4	_fp_hw	
23	-----	0x00004010	GLOBAL	NOTYPE	0	__bss_start	
24	0x00001215	0x00001215	GLOBAL	FUNC	53	main	
25	0x0000124a	0x0000124a	GLOBAL	FUNC	0	__x86.get_pc_thunk.ax	
26	-----	0x00004010	GLOBAL	OBJ	0	__TMC_END__	

импорты и экспорты

nth	paddr	size	vaddr	vsize	perm	entropy	type	name
0	0x00000000	0x0	0x00000000	0x0	----		NULL	
1	0x000001b4	0x13	0x000001b4	0x13	-r--	3.68188080	PROGBITS	.interp
2	0x000001c8	0x24	0x000001c8	0x24	-r--	4.24716720	NOTE	.note.gnu.build-id
3	0x000001ec	0x34	0x000001ec	0x34	-r--	2.11650649	NOTE	.note.gnu.property
4	0x00000220	0x20	0x00000220	0x20	-r--	1.41268789	NOTE	.note.ABI-tag
5	0x00000240	0x20	0x00000240	0x20	-r--	1.99254169	GNU_HASH	.gnu.hash
6	0x00000260	0x80	0x00000260	0x80	-r--	1.04726900	DYNSYM	.dynsym
7	0x000002e0	0xa6	0x000002e0	0xa6	-r--	4.70680756	STRTAB	.dynstr
8	0x00000386	0x10	0x00000386	0x10	-r--	1.71691719	GNU_VERSION	.gnu.version
9	0x00000398	0x40	0x00000398	0x40	-r--	2.44324982	GNU_VERNEED	.gnu.version_r
10	0x000003d8	0x40	0x000003d8	0x40	-r--	2.51036592	REL	.rel.dyn
11	0x00000418	0x10	0x00000418	0x10	-r--	1.84191719	REL	.rel.plt
12	0x00001000	0x24	0x00001000	0x24	-r-x	4.29187943	PROGBITS	.init
13	0x00001030	0x30	0x00001030	0x30	-r-x	2.50129389	PROGBITS	.plt
14	0x00001060	0x1ee	0x00001060	0x1ee	-r-x	5.65184675	PROGBITS	.text
15	0x00001250	0x18	0x00001250	0x18	-r-x	4.16829583	PROGBITS	.fini
16	0x00002000	0x27	0x00002000	0x27	-r--	3.73403059	PROGBITS	.rodata
17	0x00002028	0x34	0x00002028	0x34	-r--	3.21086382	PROGBITS	.eh_frame_hdr
18	0x0000205c	0xb8	0x0000205c	0xb8	-r--	4.25929376	PROGBITS	.eh_frame
19	0x00002ee8	0x4	0x00003ee8	0x4	-rw-	1.50000000	INIT_ARRAY	.init_array
20	0x00002eec	0x4	0x00003eec	0x4	-rw-	1.50000000	FINI_ARRAY	.fini_array
21	0x00002ef0	0xf0	0x00003ef0	0xf0	-rw-	2.53047429	DYNAMIC	.dynamic
22	0x00002fe0	0x14	0x00003fe0	0x14	-rw-	0.56899559	PROGBITS	.got
23	0x00002ff4	0x14	0x00003ff4	0x14	-rw-	1.55677965	PROGBITS	.got.plt
24	0x00003008	0x8	0x00004008	0x8	-rw-	1.06127812	PROGBITS	.data
25	0x00003010	0x0	0x00004010	0x4	-rw-		NOBITS	.bss
26	0x00003010	0x1b	0x00000000	0x1b	----	3.75416343	PROGBITS	.comment

энтропия

GCC32~01

Ghidra

```
int main (char **argv);
; var int32_t var_4h @ ebp-0x4
; arg char **argv @ esp+0x24
0x00001215    lea ecx, [argv]
0x00001219    and esp, 0xffffffff0
0x0000121c    push dword [ecx - 4]
0x0000121f    push ebp
0x00001220    mov ebp, esp
0x00001222    push ecx
0x00001223    sub esp, 4
0x00001226    call sym.__x86.get_pc_thunk.ax
0x0000122b    add eax, 0x2dc9
0x00001230    sub esp, 0xc
0x00001233    push 2
0x00001235    call sym.f
0x0000123a    add esp, 0x10
0x0000123d    mov eax, 0
0x00001242    mov ecx, dword [var_4h]
0x00001245    leave
0x00001246    lea esp, [ecx - 4]
0x00001249    ret
```

дизасемблирование main

```
4 int main(void)
5
6 {
7     f(2);
8     return 0;
9 }
```

декомпилирование


```

; CALL XREF from main @ 0x1235(x)
136: sym.f (signed int arg_8h);
; arg signed int arg_8h @ ebp+0x8
; var int32_t var_4h @ ebp-0x4
0x0000118d    push ebp
0x0000118e    mov  ebp, esp
0x00001190    push ebx
0x00001191    sub  esp, 4
0x00001194    call sym.__x86.get_pc_thunk.ax
0x00001199    add  eax, 0x2e5b
0x0000119e    cmp  dword [arg_8h], 2
0x000011a2    je   0x11e4
0x000011a4    cmp  dword [arg_8h], 2
0x000011a8    jg   0x11fa
0x000011aa    cmp  dword [arg_8h], 0
0x000011ae    je   0x11b8
0x000011b0    cmp  dword [arg_8h], 1
0x000011b4    je   0x11ce
0x000011b6    jmp  0x11fa
; CODE XREF from sym.f @ 0x11ae(x)
0x000011b8    sub  esp, 0xc
0x000011bb    lea  edx, [eax - 0x1fec]
0x000011c1    push edx
0x000011c2    mov  ebx, eax
0x000011c4    call sym.imp.puts
0x000011c9    add  esp, 0x10
0x000011cc    jmp  0x120f
; CODE XREF from sym.f @ 0x11b4(x)
0x000011ce    sub  esp, 0xc
0x000011d1    lea  edx, [eax - 0x1fe7]
0x000011d7    push edx
0x000011d8    mov  ebx, eax
0x000011da    call sym.imp.puts
0x000011df    add  esp, 0x10
0x000011e2    jmp  0x120f
; CODE XREF from sym.f @ 0x11a2(x)

```

функция
f

```

4 void f(int value)
5
6 {
7     if (value == 2) {
8         puts("two");
9         return;
10    }
11    if (value < 3) {
12        if (value == 0) {
13            puts("zero");
14            return;
15        }
16        if (value == 1) {
17            puts("one");
18            return;
19        }
20    }
21    puts("something unknown");
22    return;
23 }

```

GCC32~03

Radare2

nth	paddr	size	vaddr	vsize	perm	entropy	type	name
0	0x00000000	0x0	0x00000000	0x0	----		NULL	
1	0x000001b4	0x13	0x000001b4	0x13	-r--	3.68188080	PROGBITS	.interp
2	0x000001c8	0x24	0x000001c8	0x24	-r--	4.08050053	NOTE	.note.gnu.build-id
3	0x000001ec	0x34	0x000001ec	0x34	-r--	2.11650649	NOTE	.note.gnu.property
4	0x00000220	0x20	0x00000220	0x20	-r--	1.41268789	NOTE	.note.ABI-tag
5	0x00000240	0x20	0x00000240	0x20	-r--	1.99254169	GNU_HASH	.gnu.hash
6	0x00000260	0x80	0x00000260	0x80	-r--	1.04726900	DYNSYM	.dynsym
7	0x000002e0	0xa6	0x000002e0	0xa6	-r--	4.70680756	STRTAB	.dynstr
8	0x00000386	0x10	0x00000386	0x10	-r--	1.71691719	GNU_VERSION	.gnu.version
9	0x00000398	0x40	0x00000398	0x40	-r--	2.44324982	GNU_VERNEED	.gnu.version_r
10	0x000003d8	0x40	0x000003d8	0x40	-r--	2.51036592	REL	.rel.dyn
11	0x00000418	0x10	0x00000418	0x10	-r--	1.84191719	REL	.rel.plt
12	0x00001000	0x24	0x00001000	0x24	-r-x	4.30549679	PROGBITS	.init
13	0x00001030	0x30	0x00001030	0x30	-r-x	2.50129389	PROGBITS	.plt
14	0x00001060	0x207	0x00001060	0x207	-r-x	5.53875001	PROGBITS	.text
15	0x00001268	0x18	0x00001268	0x18	-r-x	4.08496250	PROGBITS	.fini
16	0x00002000	0x27	0x00002000	0x27	-r--	3.73403059	PROGBITS	.rodata
17	0x00002028	0x2c	0x00002028	0x2c	-r--	3.10003304	PROGBITS	.eh_frame_hdr
18	0x00002054	0xec	0x00002054	0xec	-r--	4.73878819	PROGBITS	.eh_frame
19	0x00002ee8	0x4	0x00003ee8	0x4	-rw-	1.50000000	INIT_ARRAY	.init_array
20	0x00002eec	0x4	0x00003eec	0x4	-rw-	1.50000000	FINI_ARRAY	.fini_array
21	0x00002ef0	0xf0	0x00003ef0	0xf0	-rw-	2.53047429	DYNAMIC	.dynamic
22	0x00002fe0	0x14	0x00003fe0	0x14	-rw-	0.56899559	PROGBITS	.got
23	0x00002ff4	0x14	0x00003ff4	0x14	-rw-	1.55677965	PROGBITS	.got.plt
24	0x00003008	0x8	0x00004008	0x8	-rw-	1.06127812	PROGBITS	.data
25	0x00003010	0x0	0x00004010	0x4	-rw-		NOBITS	.bss
26	0x00003010	0x1b	0x00000000	0x1b	----	3.75416343	PROGBITS	.comment
27	0x0000302c	0x1c0	0x00000000	0x1c0	----	2.32614519	SYMTAB	.symtab
28	0x000031ec	0x159	0x00000000	0x159	----	4.92806096	STRTAB	.strtab
29	0x00003345	0x114	0x00000000	0x114	----	4.26268756	STRTAB	.shstrtab

импорты и экспорты

[Imports]							
nth	vaddr	bind	type	lib name			
1	0x00001040	GLOBAL	FUNC	__libc_start_main			
2	-----	WEAK	NOTYPE	_ITM_deregisterTMCloneTable			
3	-----	WEAK	FUNC	__cxa_finalize			
4	0x00001050	GLOBAL	FUNC	puts			
5	-----	WEAK	NOTYPE	__gmon_start__			
6	-----	WEAK	NOTYPE	_ITM_registerTMCloneTable			
[Exports]							
nth	paddr	vaddr	bind	type	size	lib name	demangled
7	0x00002004	0x00002004	GLOBAL	OBJ	4	_IO_stdin_used	
8	0x000010d0	0x000010d0	GLOBAL	FUNC	4	__x86.get_pc_thunk.bx	
10	-----	0x00004010	GLOBAL	NOTYPE	0	_edata	
11	0x00001268	0x00001268	GLOBAL	FUNC	0	_fini	
12	0x000011d0	0x000011d0	GLOBAL	FUNC	151	f	
13	0x000011c9	0x000011c9	GLOBAL	FUNC	0	__x86.get_pc_thunk.dx	
15	0x00003008	0x00004008	GLOBAL	NOTYPE	0	__data_start	
18	0x0000300c	0x0000400c	GLOBAL	OBJ	0	__dso_handle	
19	0x00002004	0x00002004	GLOBAL	OBJ	4	_IO_stdin_used	
20	-----	0x00004014	GLOBAL	NOTYPE	0	_end	
21	0x000010a0	0x000010a0	GLOBAL	FUNC	48	_start	
22	0x00002000	0x00002000	GLOBAL	OBJ	4	_fp_hw	
23	-----	0x00004010	GLOBAL	NOTYPE	0	__bss_start	
24	0x00001060	0x00001060	GLOBAL	FUNC	56	main	
25	-----	0x00004010	GLOBAL	OBJ	0	__TMC_END__	
27	0x00001000	0x00001000	GLOBAL	FUNC	0	init	

энтропия

GCC32~03

Ghidra

```
    ;-- section..text:
int main (char **argv);
    ; var int32_t var_8h @ ebp-0x8
    ; arg char **argv @ esp+0x24
0x00001060    lea ecx, [argv]
0x00001064    and esp, 0xffffffff0
0x00001067    push dword [ecx - 4]
0x0000106a    push ebp
0x0000106b    mov ebp, esp
0x0000106d    push ebx
0x0000106e    call sym.__x86.get_pc_thunk.bx
0x00001073    add ebx, 0x2f81
0x00001079    push ecx
0x0000107a    sub esp, 0xc
0x0000107d    lea eax, [ebx - 0x1fe3]
0x00001083    push eax
0x00001084    call sym.imp.puts
0x00001089    add esp, 0x10
0x0000108c    lea esp, [var_8h]
0x0000108f    xor eax, eax
0x00001091    pop ecx
0x00001092    pop ebx
0x00001093    pop ebp
0x00001094    lea esp, [ecx - 4]
0x00001097    ret
```

дизасемблирование main

```
4 int main(void)
5
6 {
7     puts("two");
8     return 0;
9 }
```

декомпилирование (да, всё так - не ошибка)

```

sym.f (int32_t arg_10h);
; arg int32_t arg_10h @ esp+0x20
0x000011d0      push ebx
0x000011d1      call sym.__x86.get_pc_thunk.bx
0x000011d6      add ebx, 0x2e1e
0x000011dc      sub esp, 8
0x000011df      mov eax, dword [arg_10h]
0x000011e3      cmp eax, 1
0x000011e6      je 0x1230
0x000011e8      cmp eax, 2
0x000011eb      je 0x1210
0x000011ed      test eax, eax
0x000011ef      jne 0x1250
0x000011f1      sub esp, 0xc
0x000011f4      lea eax, [ebx - 0x1fec]
0x000011fa      push eax
0x000011fb      call sym.imp.puts
0x00001200      add esp, 0x10
0x00001203      add esp, 8
0x00001206      pop ebx
0x00001207      ret

; CODE XREF from sym.f @ 0x11eb(x)
0x00001210      sub esp, 0xc
0x00001213      lea eax, [ebx - 0x1fe3]
0x00001219      push eax
0x0000121a      call sym.imp.puts
0x0000121f      add esp, 0x10
0x00001222      add esp, 8
0x00001225      pop ebx
0x00001226      ret

; CODE XREF from sym.f @ 0x11e6(x)
0x00001230      sub esp, 0xc
0x00001233      lea eax, [ebx - 0x1fe7]
0x00001239      push eax

```

функция
f

```

4 void f(int value)
5
6 {
7     if (value == 2) {
8         puts("two");
9         return;
10    }
11    if (value < 3) {
12        if (value == 0) {
13            puts("zero");
14            return;
15        }
16        if (value == 1) {
17            puts("one");
18            return;
19        }
20    }
21    puts("something unknown");
22    return;
23 }

```


GCC64~01

Radare2

[Imports]							
nth	vaddr	bind	type	lib name			
1	-----	GLOBAL	FUNC	__libc_start_main			
2	-----	WEAK	NOTYPE	_ITM_deregisterTMCloneTable			
3	0x00001030	GLOBAL	FUNC	puts			
4	-----	WEAK	NOTYPE	__gmon_start__			
5	-----	WEAK	NOTYPE	_ITM_registerTMCloneTable			
6	-----	WEAK	FUNC	__cxa_finalize			
[Exports]							
nth	paddr	vaddr	bind	type	size	lib name	demangled
10	-----	0x00004018	GLOBAL	NOTYPE	0	__edata	
11	0x000011bc	0x000011bc	GLOBAL	FUNC	0	__fini	
12	0x00001139	0x00001139	GLOBAL	FUNC	107	f	
13	0x00003008	0x00004008	GLOBAL	NOTYPE	0	__data_start	
15	0x00003010	0x00004010	GLOBAL	OBJ	0	__dso_handle	
16	0x00002000	0x00002000	GLOBAL	OBJ	4	_IO_stdin_used	
17	-----	0x00004020	GLOBAL	NOTYPE	0	__end	
18	0x00001040	0x00001040	GLOBAL	FUNC	38	__start	
19	-----	0x00004018	GLOBAL	NOTYPE	0	__bss_start	
20	0x000011a4	0x000011a4	GLOBAL	FUNC	21	main	
21	-----	0x00004018	GLOBAL	OBJ	0	__TMC_END__	
24	0x00001000	0x00001000	GLOBAL	FUNC	0	__init	

импорты и экспорты

nth	paddr	size	vaddr	vsize	perm	entropy	type	name
0	0x00000000	0x0	0x00000000	0x0	----		NULL	
1	0x00000318	0x1c	0x00000318	0x1c	-r--	3.94075983	PROGBITS	.interp
2	0x00000338	0x40	0x00000338	0x40	-r--	1.83469501	NOTE	.note.gnu.property
3	0x00000378	0x24	0x00000378	0x24	-r--	4.19161164	NOTE	.note.gnu.build-id
4	0x0000039c	0x20	0x0000039c	0x20	-r--	1.41268789	NOTE	.note.ABI-tag
5	0x000003c0	0x1c	0x000003c0	0x1c	-r--	0.49123734	GNU_HASH	.gnu.hash
6	0x000003e0	0xa8	0x000003e0	0xa8	-r--	0.57518865	DYNSYM	.dynsym
7	0x00000488	0x8d	0x00000488	0x8d	-r--	4.64227179	STRTAB	.dynstr
8	0x00000516	0xe	0x00000516	0xe	-r--	1.61057724	GNU_VERSYM	.gnu.version
9	0x00000528	0x30	0x00000528	0x30	-r--	2.30204405	GNU_VERNEED	.gnu.version_r
10	0x00000558	0xc0	0x00000558	0xc0	-r--	1.37722996	RELA	.rela.dyn
11	0x00000618	0x18	0x00000618	0x18	-r--	0.74168476	RELA	.rela.plt
12	0x00001000	0x1b	0x00001000	0x1b	-r-x	4.23636898	PROGBITS	.init
13	0x00001020	0x20	0x00001020	0x20	-r-x	3.05254968	PROGBITS	.plt
14	0x00001040	0x179	0x00001040	0x179	-r-x	5.41642214	PROGBITS	.text
15	0x000011bc	0xd	0x000011bc	0xd	-r-x	3.23890126	PROGBITS	.fini
16	0x00002000	0x23	0x00002000	0x23	-r--	3.77259892	PROGBITS	.rodata
17	0x00002024	0x2c	0x00002024	0x2c	-r--	3.16264412	PROGBITS	.eh_frame_hdr
18	0x00002050	0x9c	0x00002050	0x9c	-r--	3.99167288	PROGBITS	.eh_frame
19	0x00002dd0	0x8	0x00003dd0	0x8	-rw-	1.06127812	INIT_ARRAY	.init_array
20	0x00002dd8	0x8	0x00003dd8	0x8	-rw-	1.06127812	FINI_ARRAY	.fini_array
21	0x00002de0	0x1e0	0x00003de0	0x1e0	-rw-	1.44803204	DYNAMIC	.dynamic
22	0x00002fc0	0x28	0x00003fc0	0x28	-rw-	0.00000000	PROGBITS	.got
23	0x00002fe8	0x20	0x00003fe8	0x20	-rw-	0.79356444	PROGBITS	.got.plt
24	0x00003008	0x10	0x00004008	0x10	-rw-	0.66856444	PROGBITS	.data
25	0x00003018	0x0	0x00004018	0x8	-rw-		NOBITS	.bss
26	0x00003018	0x1b	0x00000000	0x1b	----	3.75416343	PROGBITS	.comment
27	0x00003038	0x258	0x00000000	0x258	----	1.55603422	SYMTAB	.symtab
28	0x00003290	0x128	0x00000000	0x128	----	4.81904706	STRTAB	.strtab
29	0x000033b8	0x116	0x00000000	0x116	----	4.26099275	STRTAB	.shstrtab

энтропия

GCC64~01

Ghidra

```
21: int main (int argc, char **argv, char **envp);  
    0x000011a4    push rbp  
    0x000011a5    mov  rbp, rsp  
    0x000011a8    mov  edi, 2  
    0x000011ad    call sym.f  
    0x000011b2    mov  eax, 0  
    0x000011b7    pop  rbp  
    0x000011b8    ret
```

дизасемблирование main

```
2 int main(void)  
3  
4 {  
5     f(2);  
6     return 0;  
7 }
```

декомпилирование


```

; CALL XREF from main @ 0x11ad(x)
107: sym.f (signed int arg1);
; arg signed int arg1 @ rdi
; var signed int var_4h @ rbp-0x4
0x00001139    push rbp
0x0000113a    mov rbp, rsp
0x0000113d    sub rsp, 0x10
0x00001141    mov dword [var_4h], edi
0x00001144    cmp dword [var_4h], 2
0x00001148    je 0x1180
0x0000114a    cmp dword [var_4h], 2
0x0000114e    jg 0x1191
0x00001150    cmp dword [var_4h], 0
0x00001154    je 0x115e
0x00001156    cmp dword [var_4h], 1
0x0000115a    je 0x116f
0x0000115c    jmp 0x1191
; CODE XREF from sym.f @ 0x1154(x)
0x0000115e    lea rax, str.zero
0x00001165    mov rdi, rax
0x00001168    call sym.imp.puts
0x0000116d    jmp 0x11a1
; CODE XREF from sym.f @ 0x115a(x)
0x0000116f    lea rax, [0x00002009]
0x00001176    mov rdi, rax
0x00001179    call sym.imp.puts
0x0000117e    jmp 0x11a1
; CODE XREF from sym.f @ 0x1148(x)
0x00001180    lea rax, [0x0000200d]
0x00001187    mov rdi, rax
0x0000118a    call sym.imp.puts
0x0000118f    jmp 0x11a1
; CODE XREFS from sym.f @ 0x114e(x), 0x115c(x)
0x00001191    lea rax, str.something_unknown
0x00001198    mov rdi, rax
0x0000119b    call sym.imp.puts

```

.
 функция
 f

```

4 void f(int value)
5
6 {
7     if (value == 2) {
8         puts("two");
9         return;
10    }
11    if (value < 3) {
12        if (value == 0) {
13            puts("zero");
14            return;
15        }
16        if (value == 1) {
17            puts("one");
18            return;
19        }
20    }
21    puts("something unknown");
22    return;
23 }

```

GCC64~03

Radare2

[Imports]				
nth	vaddr	bind	type	lib name
1	-----	GLOBAL	FUNC	__libc_start_main
2	-----	WEAK	NOTYPE	__ITM_deregisterTMCloneTable
3	0x00001030	GLOBAL	FUNC	puts
4	-----	WEAK	NOTYPE	__gmon_start__
5	-----	WEAK	NOTYPE	__ITM_registerTMCloneTable
6	-----	WEAK	FUNC	__cxa_finalize

[Exports]						
nth	paddr	vaddr	bind	type	size lib name	demangled
10	-----	0x00004018	GLOBAL	NOTYPE	0	__edata
11	0x000011ac	0x000011ac	GLOBAL	FUNC	0	__fini
12	0x00001160	0x00001160	GLOBAL	FUNC	76	f
13	0x00003008	0x00004008	GLOBAL	NOTYPE	0	__data_start
15	0x00003010	0x00004010	GLOBAL	OBJ	0	__dso_handle
16	0x00002000	0x00002000	GLOBAL	OBJ	4	__IO_stdin_used
17	-----	0x00004020	GLOBAL	NOTYPE	0	__end
18	0x00001060	0x00001060	GLOBAL	FUNC	38	__start
19	-----	0x00004018	GLOBAL	NOTYPE	0	__bss_start
20	0x00001040	0x00001040	GLOBAL	FUNC	23	main
21	-----	0x00004018	GLOBAL	OBJ	0	__TMC_END__
24	0x00001000	0x00001000	GLOBAL	FUNC	0	__init

импорты и экспорты

nth	paddr	size	vaddr	vsize	perm	entropy	type	name
0	0x00000000	0x0	0x00000000	0x0	----		NULL	
1	0x00000318	0x1c	0x00000318	0x1c	-r--	3.94075983	PROGBITS	.interp
2	0x00000338	0x40	0x00000338	0x40	-r--	1.83469501	NOTE	.note.gnu.property
3	0x00000378	0x24	0x00000378	0x24	-r--	4.13605609	NOTE	.note.gnu.build-id
4	0x0000039c	0x20	0x0000039c	0x20	-r--	1.41268789	NOTE	.note.ABI-tag
5	0x000003c0	0x1c	0x000003c0	0x1c	-r--	0.49123734	GNU_HASH	.gnu.hash
6	0x000003e0	0xa8	0x000003e0	0xa8	-r--	0.57518865	DYNSYM	.dynsym
7	0x00000488	0x8d	0x00000488	0x8d	-r--	4.64227179	STRTAB	.dynstr
8	0x00000516	0xe	0x00000516	0xe	-r--	1.61057724	GNU_VERSION	.gnu.version
9	0x00000528	0x30	0x00000528	0x30	-r--	2.30204405	GNU_VERSION	.gnu.version_r
10	0x00000558	0xc0	0x00000558	0xc0	-r--	1.34604760	RELA	.rela.dyn
11	0x00000618	0x18	0x00000618	0x18	-r--	0.74168476	RELA	.rela.plt
12	0x00001000	0x1b	0x00001000	0x1b	-r-x	4.23636898	PROGBITS	.init
13	0x00001020	0x20	0x00001020	0x20	-r-x	3.05254968	PROGBITS	.plt
14	0x00001040	0x16c	0x00001040	0x16c	-r-x	5.13468026	PROGBITS	.text
15	0x000011ac	0xd	0x000011ac	0xd	-r-x	3.23890126	PROGBITS	.fini
16	0x00002000	0x23	0x00002000	0x23	-r--	3.77259892	PROGBITS	.rodata
17	0x00002024	0x2c	0x00002024	0x2c	-r--	3.11718957	PROGBITS	.eh_frame_hdr
18	0x00002050	0x88	0x00002050	0x88	-r--	3.54112256	PROGBITS	.eh_frame
19	0x00002dd0	0x8	0x00003dd0	0x8	-rw-	1.06127812	INIT_ARRAY	.init_array
20	0x00002dd8	0x8	0x00003dd8	0x8	-rw-	0.54356444	FINI_ARRAY	.fini_array
21	0x00002de0	0x1e0	0x00003de0	0x1e0	-rw-	1.44803204	DYNAMIC	.dynamic
22	0x00002fc0	0x28	0x00003fc0	0x28	-rw-	0.00000000	PROGBITS	.got
23	0x00002fe8	0x20	0x00003fe8	0x20	-rw-	0.79356444	PROGBITS	.got.plt
24	0x00003008	0x10	0x00004008	0x10	-rw-	0.66856444	PROGBITS	.data
25	0x00003018	0x0	0x00004018	0x0	-rw-		NOBITS	.bss
26	0x00003018	0x1b	0x00000000	0x1b	----	3.75416343	PROGBITS	.comment
27	0x00003038	0x258	0x00000000	0x258	----	1.55573287	SYMTAB	.symtab
28	0x00003290	0x128	0x00000000	0x128	----	4.81904706	STRTAB	.strtab
29	0x000033b8	0x116	0x00000000	0x116	----	4.26099275	STRTAB	.shstrtab

энтропия

GCC64~03

Ghidra

```
int main (int argc, char **argv, char **envp);  
    0x00001040    sub rsp, 8  
    0x00001044    lea rdi, [0x0000200d]  
    0x0000104b    call sym.imp.puts  
    0x00001050    xor eax, eax  
    0x00001052    add rsp, 8  
    0x00001056    ret
```

дизасемблирование main

```
2 int main(void)  
3  
4 {  
5     puts("two");  
6     return 0;  
7 }
```

декомпилирование

```

62: sym.f (uint32_t arg1);
    ; arg uint32_t arg1 @ rdi
    0x00001160    cmp edi, 1
    0x00001163    je 0x1190
    0x00001165    cmp edi, 2
    0x00001168    je 0x1180
    0x0000116a    test edi, edi
    0x0000116c    jne 0x11a0
    0x0000116e    lea rdi, str.zero
    0x00001175    jmp sym.imp.puts

..
    ; CODE XREF from sym.f @ 0x1168(x)
    0x00001180    lea rdi, [0x0000200d]
    0x00001187    jmp sym.imp.puts

..
    ; CODE XREF from sym.f @ 0x1163(x)
    0x00001190    lea rdi, [0x00002009]
    0x00001197    jmp sym.imp.puts

..
    ; CODE XREF from sym.f @ 0x116c(x)
    0x000011a0    lea rdi, str.something_unknown
    0x000011a7    jmp sym.imp.puts

```

функция
f

```

2 void f(int value)
3
4 {
5     if (value == 1) {
6         puts("one");
7         return;
8     }
9     if (value != 2) {
10        if (value == 0) {
11            puts("zero");
12            return;
13        }
14        puts("something unknown");
15        return;
16    }
17    puts("two");
18    return;
19 }

```


Сравнение GCC и Clang 64bit с оптимизацией

Всего отличаются на 2066 байт

00000019 60 40
00000029 D0 E0
000000D1 30 10
000000D9 30 10
00000109 B9 91
00000111 B9 91
00000141 D8 E8
00000149 D8 E8
000001E9 40 20
000001F1 40 20
00000209 78 58
00000211 78 58

00000219 78 58
00000259 40 20
00000261 40 20
00000279 24 30
00000281 24 30
00000289 24 30
0000033D 30 10
00000359 01 04
0000035B 01 00
0000035C C0 00
0000035D 04 14
00000361 01 03

Различие по адресу 0x00000019 60 40

		18	19	1A	
GCC	0x00000017	00	60	10	add byte [rax + 0x10], ah
CLANG	0x00000017	00	40	10	add byte [rax + 0x10], al

Различие по адресу 0x00000029 d0 e0

		28	29	
GCC	0x00000027	00	60	add al, dl
CLANG	0x00000027	00	40	add al, ah

00001091 48 e0

00001092 8d 0f

00001093 3d 1f

00001094 81 80

00001095 2f 00

		91	92	93	94	95	96	
GCC	0x00001090	48	8d	3d	81	2f	00	lea rdi, loc._edata
CLANG	0x00001090	e0	0f	1f	80	00	00	nop dword [rax]

CLANG32~01 vs CLANG32~03

Ghidra

```
0x000011d0    push ebx
0x000011d1    sub esp, 8
0x000011d4    call 0x11d9
; CALL XREF from main @ 0x11d4(x)
0x000011d9    pop ebx
0x000011da    add ebx, 0x2e1b
0x000011e0    lea eax, [ebx - 0x1fec]
0x000011e6    mov dword [esp], eax
0x000011e9    call sym.imp.puts
0x000011ee    xor eax, eax
0x000011f0    add esp, 8
0x000011f3    pop ebx
0x000011f4    ret
```

CLANG32-01

```
0x000011d0    push ebx
0x000011d1    sub esp, 8
0x000011d4    call 0x11d9
; CALL XREF from main @ 0x11d4(x)
0x000011d9    pop ebx
0x000011da    add ebx, 0x2e1b
0x000011e0    lea eax, [ebx - 0x1fec]
0x000011e6    mov dword [esp], eax
0x000011e9    call sym.imp.puts
0x000011ee    xor eax, eax
0x000011f0    add esp, 8
0x000011f3    pop ebx
0x000011f4    ret
```

CLANG32-03

CLANG64~01 vs CLANG64~03

Ghidra

```
; DATA XREF from entry0 @ 0x1058(r)
main (int argc, char **argv, char **envp)
0x00001170      push rax
0x00001171      lea rdi, [0x00002004]
0x00001178      call sym.imp.puts
0x0000117d      xor eax, eax
0x0000117f      pop rcx
0x00001180      ret
```

CLANG64-01

```
; DATA XREF from entry0 @ 0x1058(r)
main (int argc, char **argv, char **envp)
0x00001170      push rax
0x00001171      lea rdi, [0x00002004]
0x00001178      call sym.imp.puts
0x0000117d      xor eax, eax
0x0000117f      pop rcx
0x00001180      ret
```

CLANG64-03

GCC32~01 vs GCC32~03

Ghidra

```
0x00001215    lea ecx, [argv]
0x00001219    and esp, 0xffffffff0
0x0000121c    push dword [ecx - 4]
0x0000121f    push ebp
0x00001220    mov ebp, esp
0x00001222    push ecx
0x00001223    sub esp, 4
0x00001226    call sym.__x86.get_pc_thunk.ax
0x0000122b    add eax, 0x2dc9
0x00001230    sub esp, 0xc
0x00001233    push 2
0x00001235    call sym.f
0x0000123a    add esp, 0x10
0x0000123d    mov eax, 0
0x00001242    mov ecx, dword [var_4h]
0x00001245    leave
0x00001246    lea esp, [ecx - 4]
0x00001249    ret
```

GCC32-01

```
0x00001060    lea ecx, [argv]
0x00001064    and esp, 0xffffffff0
0x00001067    push dword [ecx - 4]
0x0000106a    push ebp
0x0000106b    mov ebp, esp
0x0000106d    push ebx
0x0000106e    call sym.__x86.get_pc_thunk.bx
0x00001073    add ebx, 0x2f81
0x00001079    push ecx
0x0000107a    sub esp, 0xc
0x0000107d    lea eax, [ebx - 0x1fe3]
0x00001083    push eax
0x00001084    call sym.imp.puts
0x00001089    add esp, 0x10
0x0000108c    lea esp, [var_8h]
0x0000108f    xor eax, eax
0x00001091    pop ecx
0x00001092    pop ebx
0x00001093    pop ebp
0x00001094    lea esp, [ecx - 4]
0x00001097    ret
```

GCC32-03

GCC64~01 vs GCC64~03

Ghidra

```
0x000011a4    push rbp
0x000011a5    mov  rbp, rsp
0x000011a8    mov  edi, 2
0x000011ad    call sym.f
0x000011b2    mov  eax, 0
0x000011b7    pop  rbp
0x000011b8    ret
```

GCC64-01

```
0x00001040    sub  rsp, 8
0x00001044    lea  rdi, [0x0000200d]
0x0000104b    call sym.imp.puts
0x00001050    xor  eax, eax
0x00001052    add  rsp, 8
0x00001056    ret
```

GCC64-03

GCC32~01 vs CLANG32~01

Ghidra

```
0x00001215    lea ecx, [argv]
0x00001219    and esp, 0xffffffff
0x0000121c    push dword [ecx - 4]
0x0000121f    push ebp
0x00001220    mov ebp, esp
0x00001222    push ecx
0x00001223    sub esp, 4
0x00001226    call sym.__x86.get_pc_thunk.ax
0x0000122b    add eax, 0x2dc9
0x00001230    sub esp, 0xc
0x00001233    push 2
0x00001235    call sym.f
0x0000123a    add esp, 0x10
0x0000123d    mov eax, 0
0x00001242    mov ecx, dword [var_4h]
0x00001245    leave
0x00001246    lea esp, [ecx - 4]
0x00001249    ret
```

GCC32-01

```
0x000011d0    push ebx
0x000011d1    sub esp, 8
0x000011d4    call 0x11d9
; CALL XREF from main @ 0x11d4(x)
0x000011d9    pop ebx
0x000011da    add ebx, 0x2e1b
0x000011e0    lea eax, [ebx - 0x1fec]
0x000011e6    mov dword [esp], eax
0x000011e9    call sym.imp.puts
0x000011ee    xor eax, eax
0x000011f0    add esp, 8
0x000011f3    pop ebx
0x000011f4    ret
```

CLANG32-01

GCC64~01 vs CLANG64~01

Ghidra

```
0x000011a4    push rbp
0x000011a5    mov  rbp, rsp
0x000011a8    mov  edi, 2
0x000011ad    call sym.f
0x000011b2    mov  eax, 0
0x000011b7    pop  rbp
0x000011b8    ret
```

GCC64-01

```
; DATA XREF from entry0 @ 0x1058(r)
main (int argc, char **argv, char **env
0x00001170    push rax
0x00001171    lea  rdi, [0x00002004]
0x00001178    call sym.imp.puts
0x0000117d    xor  eax, eax
0x0000117f    pop  rcx
0x00001180    ret
```

CLANG64-01

Ввод с клавиатуры

Исполняемые файлы

```
1  #include <stdio.h>
2
3  void f (int a)
4  {
5      switch (a) {
6          case 0:
7              printf ("zero\n");
8              break;
9
10         case 1:
11             printf("one\n");
12             break;
13
14         case 2:
15             printf("two\n");
16             break;
17
18         default:
19             printf("something unknown\n");
20     }
21 }
22
23
24 int main(void) {
25     int input;
26     printf("Enter a number: ");
27     scanf("%d", &input);
28     f(input);
29     return 0;
30 }
```

```
test_x86_64_Scanf
test_x86_64_Scanf_clang
```


GCC64~01

Radare2

[Imports]				
nth	vaddr	bind	type	lib name
1	-----	GLOBAL	FUNC	__libc_start_main
2	-----	WEAK	NOTYPE	__ITM_deregisterTMCloneTable
3	0x00001030	GLOBAL	FUNC	puts
4	0x00001040	GLOBAL	FUNC	__stack_chk_fail
5	0x00001050	GLOBAL	FUNC	printf
6	-----	WEAK	NOTYPE	__gmon_start__
7	0x00001060	GLOBAL	FUNC	__isoc99_scanf
8	-----	WEAK	NOTYPE	__ITM_registerTMCloneTable
9	-----	WEAK	FUNC	__cxa_finalize

[Exports]						
nth	paddr	vaddr	bind	type	size	lib name
10	-----	0x00004030	GLOBAL	NOTYPE	0	__edata
11	0x00001214	0x00001214	GLOBAL	FUNC	0	__fini
13	0x00001169	0x00001169	GLOBAL	FUNC	77	f
15	0x00003020	0x00004020	GLOBAL	NOTYPE	0	__data_start
17	0x00003028	0x00004028	GLOBAL	OBJ	0	__dso_handle
18	0x00002000	0x00002000	GLOBAL	OBJ	4	__IO_stdin_used
19	-----	0x00004038	GLOBAL	NOTYPE	0	__end
20	0x00001070	0x00001070	GLOBAL	FUNC	38	__start
21	-----	0x00004030	GLOBAL	NOTYPE	0	__bss_start
22	0x000011b6	0x000011b6	GLOBAL	FUNC	94	main
24	-----	0x00004030	GLOBAL	OBJ	0	__TMC_END__
27	0x00001000	0x00001000	GLOBAL	FUNC	0	__init

импорты и экспорты

nth	paddr	size	vaddr	vsize	perm	entropy	type	name
0	0x00000000	0x0	0x00000000	0x0	----		NULL	
1	0x00000318	0x1c	0x00000318	0x1c	-r--	3.94075983	PROGBITS	.interp
2	0x00000338	0x40	0x00000338	0x40	-r--	1.83469501	NOTE	.note.gnu.property
3	0x00000378	0x24	0x00000378	0x24	-r--	4.19161164	NOTE	.note.gnu.build-id
4	0x0000039c	0x20	0x0000039c	0x20	-r--	1.41268789	NOTE	.note.ABI-tag
5	0x000003c0	0x1c	0x000003c0	0x1c	-r--	0.49123734	GNU_HASH	.gnu.hash
6	0x000003e0	0xf0	0x000003e0	0xf0	-r--	0.62053705	DYNSYM	.dynsym
7	0x000004d0	0xc8	0x000004d0	0xc8	-r--	4.74979996	STRTAB	.dynstr
8	0x00000598	0x14	0x00000598	0x14	-r--	1.94375195	GNU_VERSYM	.gnu.version
9	0x000005b0	0x50	0x000005b0	0x50	-r--	2.58779781	GNU_VERNEED	.gnu.version_r
10	0x00000600	0xc0	0x00000600	0xc0	-r--	1.37467670	RELA	.rela.dyn
11	0x000006c0	0x60	0x000006c0	0x60	-r--	1.03144697	RELA	.rela.plt
12	0x00001000	0x1b	0x00001000	0x1b	-r-x	4.23636898	PROGBITS	.init
13	0x00001020	0x50	0x00001020	0x50	-r-x	3.29205937	PROGBITS	.plt
14	0x00001070	0x1a4	0x00001070	0x1a4	-r-x	5.35232933	PROGBITS	.text
15	0x00001214	0xd	0x00001214	0xd	-r-x	3.23890126	PROGBITS	.fini
16	0x00002000	0x37	0x00002000	0x37	-r--	4.13807716	PROGBITS	.rodata
17	0x00002038	0x2c	0x00002038	0x2c	-r--	3.16264412	PROGBITS	.eh_frame_hdr
18	0x00002068	0x94	0x00002068	0x94	-r--	3.85334222	PROGBITS	.eh_frame
19	0x00002dd0	0x8	0x00003dd0	0x8	-rw-	1.06127812	INIT_ARRAY	.init_array
20	0x00002dd8	0x8	0x00003dd8	0x8	-rw-	1.06127812	FINI_ARRAY	.fini_array
21	0x00002de0	0x1e0	0x00003de0	0x1e0	-rw-	1.42644829	DYNAMIC	.dynamic
22	0x00002fc0	0x28	0x00003fc0	0x28	-rw-	0.00000000	PROGBITS	.got
23	0x00002fe8	0x38	0x00003fe8	0x38	-rw-	1.12728617	PROGBITS	.got.plt
24	0x00003020	0x10	0x00004020	0x10	-rw-	0.66856444	PROGBITS	.data
25	0x00003030	0x0	0x00004030	0x8	-rw-		NOBITS	.bss
26	0x00003030	0x1b	0x00000000	0x1b	----	3.75416343	PROGBITS	.comment
27	0x00003050	0x2a0	0x00000000	0x2a0	----	1.55371167	SYMTAB	.symtab
28	0x000032f0	0x16f	0x00000000	0x16f	----	4.91148175	STRTAB	.strtab
29	0x0000345f	0x116	0x00000000	0x116	----	4.26099275	STRTAB	.shstrtab

энтропия

GCC64~01

Ghidra

```
; DATA XREF from entry0 @ 0x1088(r)
int main (int argc, char **argv, char **envp);
; var int64_t var_4h @ rsp+0x4
; var int64_t var_8h @ rsp+0x8
0x000011b6      sub rsp, 0x18
0x000011ba      mov rax, qword fs:[0x28]
0x000011c3      mov qword [var_8h], rax
0x000011c8      xor eax, eax
0x000011ca      lea rdi, str.Enter_a_number:_
0x000011d1      call sym.imp.printf
0x000011d6      lea rsi, [var_4h]
0x000011db      lea rdi, [0x00002034]
0x000011e2      mov eax, 0
0x000011e7      call sym.imp.__isoc99_scanf
0x000011ec      mov edi, dword [var_4h]
0x000011f0      call sym.f
0x000011f5      mov rax, qword [var_8h]
0x000011fa      sub rax, qword fs:[0x28]
0x00001203      jne 0x120f
0x00001205      mov eax, 0
0x0000120a      add rsp, 0x18
0x0000120e      ret
; CODE XREF from main @ 0x1203(x)
0x0000120f      call sym.imp.__stack_chk_fail
```

дизасемблирование main

```
1
2 undefined8 main(void)
3
4 {
5     long in_FS_OFFSET;
6     int local_14;
7     long local_10;
8
9     local_10 = *(long *) (in_FS_OFFSET + 0x28);
10    printf("Enter a number: ");
11    __isoc99_scanf(&DAT_00102034,&local_14);
12    f(local_14);
13    if (local_10 == *(long *) (in_FS_OFFSET + 0x28)) {
14        return 0;
15    }
16
17    /* WARNING: Subroutine does not
18    __stack_chk_fail();
19 }
```

декомпилирование

```

; CALL XREF from main @ 0x11f0(x)
77: sym.f (uint32_t arg1);
; arg uint32_t arg1 @ rdi
0x00001169    sub rsp, 8
0x0000116d    cmp edi, 1
0x00001170    je 0x118c
0x00001172    cmp edi, 2
0x00001175    je 0x119a
0x00001177    test edi, edi
0x00001179    jne 0x11a8
0x0000117b    lea rdi, str.zero
0x00001182    call sym.imp.puts
; CODE XREFS from sym.f @ 0x1198(x), 0x11a6(x), 0x11b4(x)
0x00001187    add rsp, 8
0x0000118b    ret
; CODE XREF from sym.f @ 0x1170(x)
0x0000118c    lea rdi, [0x00002009]
0x00001193    call sym.imp.puts
0x00001198    jmp 0x1187
; CODE XREF from sym.f @ 0x1175(x)
0x0000119a    lea rdi, [0x0000200d]
0x000011a1    call sym.imp.puts
0x000011a6    jmp 0x1187
; CODE XREF from sym.f @ 0x1179(x)
0x000011a8    lea rdi, str.something_unknown
0x000011af    call sym.imp.puts
0x000011b4    jmp 0x1187

```

функция
f

```

void f(int value)
{
    if (value == 1) {
        puts("one");
    }
    else if (value == 2) {
        puts("two");
    }
    else if (value == 0) {
        puts("zero");
    }
    else {
        puts("something unknown");
    }
    return;
}

```


CLANG64~01

Radare2

[Imports]							
nth	vaddr	bind	type	lib name			
1	-----	GLOBAL	FUNC	__libc_start_main			
2	-----	WEAK	NOTYPE	_ITM_deregisterTMCloneTable			
3	0x00001030	GLOBAL	FUNC	puts			
4	0x00001040	GLOBAL	FUNC	__stack_chk_fail			
5	0x00001050	GLOBAL	FUNC	printf			
6	-----	WEAK	NOTYPE	__gmon_start__			
7	0x00001060	GLOBAL	FUNC	__isoc99_scanf			
8	-----	WEAK	NOTYPE	_ITM_registerTMCloneTable			
9	-----	WEAK	FUNC	__cxa_finalize			
[Exports]							
nth	paddr	vaddr	bind	type	size	lib name	demangled
10	-----	0x00004030	GLOBAL	NOTYPE	0	_edata	
11	0x00001214	0x00001214	GLOBAL	FUNC	0	_fini	
13	0x00001170	0x00001170	GLOBAL	FUNC	36	f	
15	0x00003020	0x00004020	GLOBAL	NOTYPE	0	__data_start	
17	0x00003028	0x00004028	GLOBAL	OBJ	0	__dso_handle	
18	0x00002000	0x00002000	GLOBAL	OBJ	4	_IO_stdin_used	
19	-----	0x00004038	GLOBAL	NOTYPE	0	_end	
20	0x00001070	0x00001070	GLOBAL	FUNC	38	_start	
21	-----	0x00004030	GLOBAL	NOTYPE	0	__bss_start	
22	0x000011a0	0x000011a0	GLOBAL	FUNC	115	main	
24	-----	0x00004030	GLOBAL	OBJ	0	__TMC_END__	
27	0x00001000	0x00001000	GLOBAL	FUNC	0	init	

импорты и экспорты

nth	paddr	size	vaddr	vsize	perm	entropy	type	name
0	0x00000000	0x0	0x00000000	0x0	----		NULL	
1	0x00000318	0x1c	0x00000318	0x1c	-r--	3.94075983	PROGBITS	.interp
2	0x00000338	0x20	0x00000338	0x20	-r--	2.05504169	NOTE	.note.gnu.property
3	0x00000358	0x24	0x00000358	0x24	-r--	4.13605609	NOTE	.note.gnu.build-id
4	0x0000037c	0x20	0x0000037c	0x20	-r--	1.41268789	NOTE	.note.ABI-tag
5	0x000003a0	0x1c	0x000003a0	0x1c	-r--	0.49123734	GNU_HASH	.gnu.hash
6	0x000003c0	0xf0	0x000003c0	0xf0	-r--	0.62053705	DYNSYM	.dynsym
7	0x000004b0	0xc8	0x000004b0	0xc8	-r--	4.74979996	STRTAB	.dynstr
8	0x00000578	0x14	0x00000578	0x14	-r--	1.94375195	GNU_VERSYM	.gnu.version
9	0x00000590	0x50	0x00000590	0x50	-r--	2.58779781	GNU_VERNEED	.gnu.version_r
10	0x000005e0	0x108	0x000005e0	0x108	-r--	1.46464232	RELA	.rela.dyn
11	0x000006e8	0x60	0x000006e8	0x60	-r--	1.03144697	RELA	.rela.plt
12	0x00001000	0x1b	0x00001000	0x1b	-r-x	4.23636898	PROGBITS	.init
13	0x00001020	0x50	0x00001020	0x50	-r-x	3.29205937	PROGBITS	.plt
14	0x00001070	0x1a3	0x00001070	0x1a3	-r-x	5.38688095	PROGBITS	.text
15	0x00001214	0xd	0x00001214	0xd	-r-x	3.23890126	PROGBITS	.fini
16	0x00002000	0x37	0x00002000	0x37	-r--	4.13807716	PROGBITS	.rodata
17	0x00002038	0x2c	0x00002038	0x2c	-r--	3.11718957	PROGBITS	.eh_frame_hdr
18	0x00002068	0x8c	0x00002068	0x8c	-r--	3.67537308	PROGBITS	.eh_frame
19	0x00002db8	0x8	0x00003db8	0x8	-rw-	1.06127812	INIT_ARRAY	.init_array
20	0x00002dc0	0x8	0x00003dc0	0x8	-rw-	1.06127812	FINI_ARRAY	.fini_array
21	0x00002dc8	0x18	0x00003dc8	0x18	-rw-	1.12418542	PROGBITS	.data.rel.ro
22	0x00002de0	0x1e0	0x00003de0	0x1e0	-rw-	1.46098548	DYNAMIC	.dynamic
23	0x00002fc0	0x28	0x00003fc0	0x28	-rw-	0.00000000	PROGBITS	.got
24	0x00002fe8	0x38	0x00003fe8	0x38	-rw-	1.12728617	PROGBITS	.got.plt
25	0x00003020	0x10	0x00004020	0x10	-rw-	0.66856444	PROGBITS	.data
26	0x00003030	0x0	0x00004030	0x8	-rw-		NOBITS	.bss
27	0x00003030	0x30	0x00000000	0x30	----	4.49403483	PROGBITS	.comment
28	0x00003060	0x2a0	0x00000000	0x2a0	----	1.54775929	SYMTAB	.symtab
29	0x00003300	0x16f	0x00000000	0x16f	----	4.91148175	STRTAB	.strtab
30	0x0000346f	0x123	0x00000000	0x123	----	4.24361117	STRTAB	.shstrtab

энтропия

CLANG64~01

Ghidra

```
115: int main (int argc, char **argv, char **envp);
    ; var int64_t var_ch @ rsp+0xc
    ; var uint32_t var_10h @ rsp+0x10
0x000011a0    sub     rsp, 0x18
0x000011a4    mov     rax, qword fs:[0x28]
0x000011ad    mov     qword [var_10h], rax
0x000011b2    lea     rdi, str.Enter_a_number:_
0x000011b9    xor     eax, eax
0x000011bb    call    sym.imp.printf
0x000011c0    lea     rdi, [0x00002015]
0x000011c7    lea     rsi, [var_ch]
0x000011cc    xor     eax, eax
0x000011ce    call    sym.imp.__isoc99_scanf
0x000011d3    movsxd  rax, dword [var_ch]
0x000011d8    cmp     rax, 2
0x000011dc    ja      0x11eb
0x000011de    lea     rcx, section..data.rel.ro
0x000011e5    mov     rdi, qword [rcx + rax*8]
0x000011e9    jmp     0x11f2
    ; CODE XREF from main @ 0x11dc(x)
0x000011eb    lea     rdi, str.something_unknown
    ; CODE XREF from main @ 0x11e9(x)
0x000011f2    call    sym.imp.puts
0x000011f7    mov     rax, qword fs:[0x28]
0x00001200    cmp     rax, qword [var_10h]
0x00001205    jne     0x120e
0x00001207    xor     eax, eax
0x00001209    add     rsp, 0x18
0x0000120d    ret
    ; CODE XREF from main @ 0x1205(x)
0x0000120e    call    sym.imp.__stack_chk_fail
```

дизасемблирование main

```
1
2 undefined8 main(void)
3
4 {
5     char *__s;
6     long in_FS_OFFSET;
7     int local_c;
8     long local_8;
9
10    local_8 = *(long *) (in_FS_OFFSET + 0x28);
11    printf("Enter a number: ");
12    __isoc99_scanf(&DAT_00102015, &local_c);
13    if ((ulong)(long)local_c < 3) {
14        __s = (&PTR_DAT_00103dc8)[local_c];
15    }
16    else {
17        __s = "something unknown";
18    }
19    puts(__s);
20    if (*(long *) (in_FS_OFFSET + 0x28) == local_8) {
21        return 0;
22    }
23
24    /* WARNING: Subroutine does not return */
25    __stack_chk_fail();
26 }
```

декомпилирование

F(a:2) vs Scanf

```
2 int main(void)
3
4 {
5     f(2);
6     return 0;
7 }
```

GCC с константой

```
1
2 undefined8 main(void)
3
4 {
5     long in_FS_OFFSET;
6     int local_14;
7     long local_10;
8
9     local_10 = *(long *) (in_FS_OFFSET + 0x28);
10    printf("Enter a number: ");
11    __isoc99_scanf(&DAT_00102034,&local_14);
12    f(local_14);
13    if (local_10 == *(long *) (in_FS_OFFSET + 0x28)) {
14        return 0;
15    }
16
17    /* WARNING: Subroutine does not
18    __stack_chk_fail();
19 }
```

GCC без константы

F(a:2) vs Scanf

```
2 int main(void)
3
4 {
5     puts("two");
6     return 0;
7 }
```

CLANG с константой

```
1
2 undefined8| main(void)
3
4 {
5     char *__s;
6     long in_FS_OFFSET;
7     int local_c;
8     long local_8;
9
10    local_8 = *(long *) (in_FS_OFFSET + 0x28);
11    printf("Enter a number: ");
12    __isoc99_scanf(&DAT_00102015,&local_c);
13    if ((ulong)(long)local_c < 3) {
14        __s = (&PTR_DAT_00103dc8)[local_c];
15    }
16    else {
17        __s = "something unknown";
18    }
19    puts(__s);
20    if (*(long *) (in_FS_OFFSET + 0x28) == local_8) {
21        return 0;
22    }
23    /* WARNING: Subroutine does not return */
24    __stack_chk_fail();
25 }
26
```

CLANG без константы

**СПАСИБО ЗА
ВНИМАНИЕ**